

## **Dev Team 42**

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## **1. INTRODUCTION**

### **1.1. Software Purpose**

The purpose of this project is to design and develop an Employee Management System (EMS) for Company Z that modernizes how employee data is handled. Instead of relying on manual MySQL scripts, this system provides a more organized, secure, and user-friendly solution. It is designed for both HR administrators and general employees, giving HR full control over managing and updating employee records, while allowing employees to easily view their own personal and payroll information. Overall, the system aims to streamline operations, improve data security, and support the company's future growth as it continues to expand.

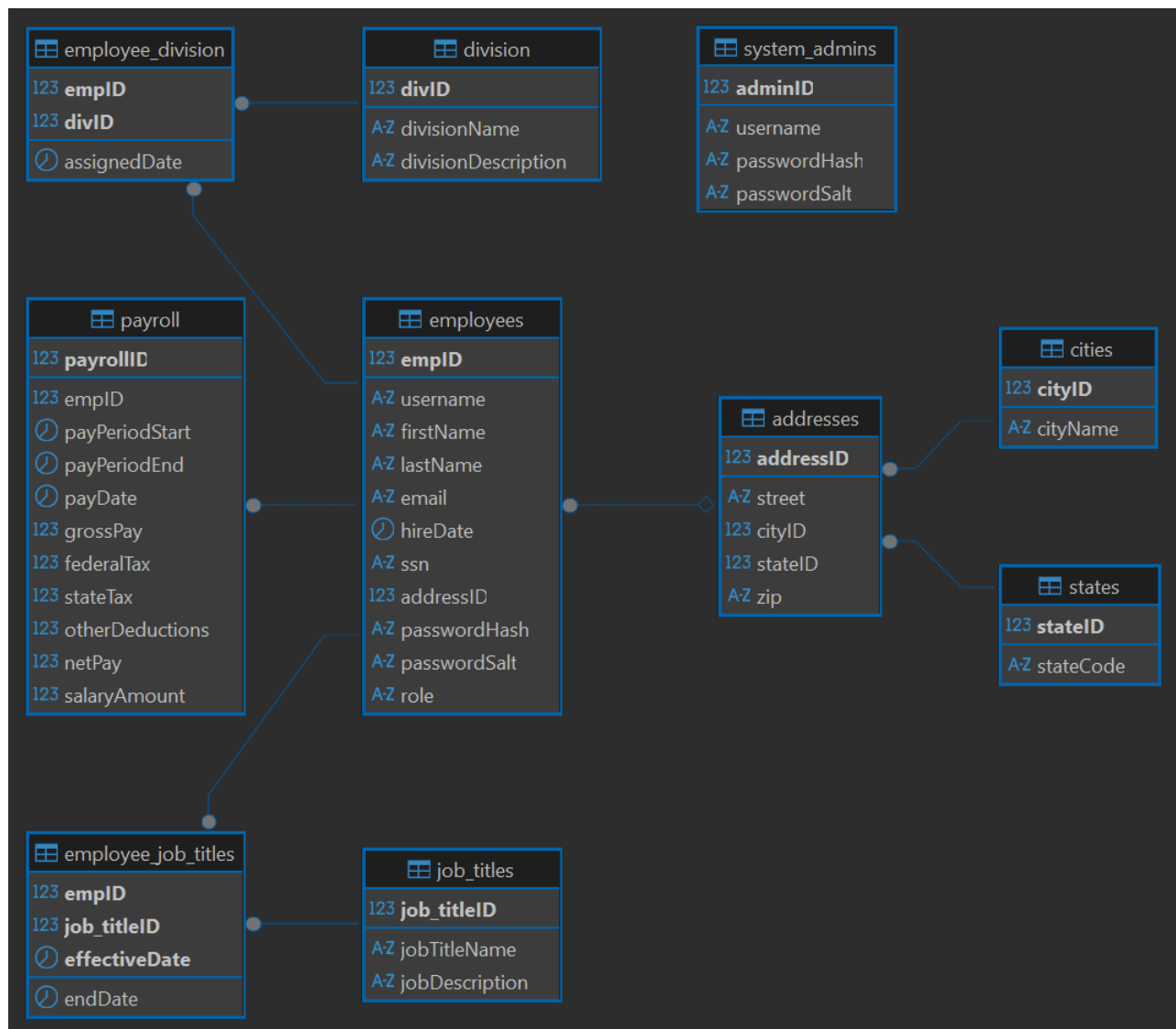
## 2. DATABASE SCHEMA DIAGRAM

This database schema is designed with MySQL and visualized using DBeaver's ER Diagram tools.

Key tables include:

- employees
- payroll
- division
- employee\_division
- job\_titles
- employee\_job\_titles
- addresses, cities, states

Relationships ensure data integrity through primary and foreign keys.



### **3. JAVA CLASS DIAGRAMS**

[Classes, interfaces, abstract classes showing details and relationships]

## **4. PROGRAMMING TASKS**

The following programming tasks were implemented:

1. Implement secure user authentication method that cross references with employee database
2. System admins can update employees' data through one interface
3. Allow users to search for their employee data using their own unique ID.
4. Search and update employees' salaries by a percentage for specific ranges of salaries.
5. Store payment history in database by date and employee ID for easy access
6. Implement role-based authorization (System admin vs general user access)
7. Develop employee deletion functionality after search (HR admin only)
8. Create pay statement history report sorted by most recent date
9. Generate monthly pay reports by job title and division
10. Implement input validation and error handling (e.g., invalid SSN, zip code, salary inputs)

## **5. TEST CASES**

### **A. Update Employee Data**

Test Case A1:

- Input: Street Name = '2167 Reggie Road'
- Expected: Update successful
- Pass: Data updates correctly
- Fail: No update

Test Case A2:

- Input: Zip Code = "BaconEggNCheese"
- Expected: Error message
- Pass: Error detected
- Fail: Invalid data accepted

### **B. Delete Employee After Search**

Test Case B1:

- Input: empID = 101, confirm YES
- Expected: Employee removed
- Pass: Employee deleted
- Fail: Employee still exists

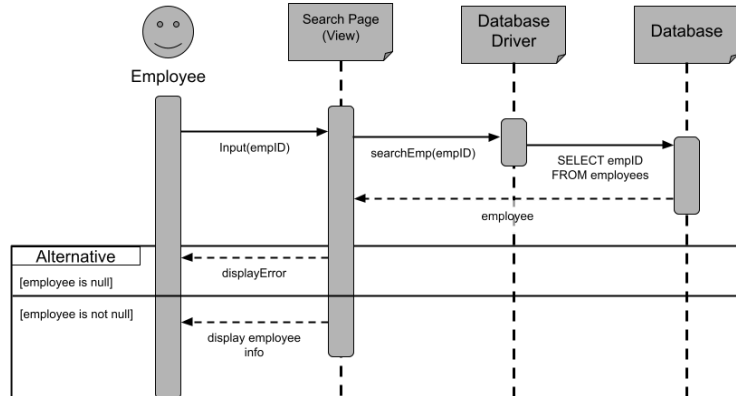
### **C. Salary Update**

Test Case C1:

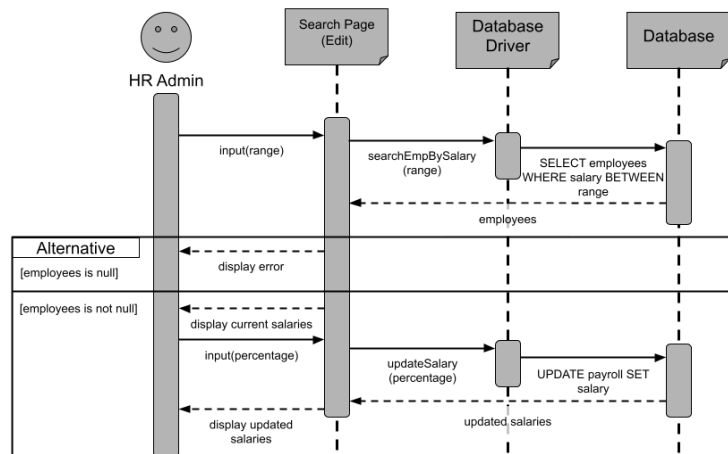
- Input: Threshold = 50000, Increase = 10%
- Expected: Only valid employees updated
- Pass: Correct updates

## 6. SEQUENCE DIAGRAMS

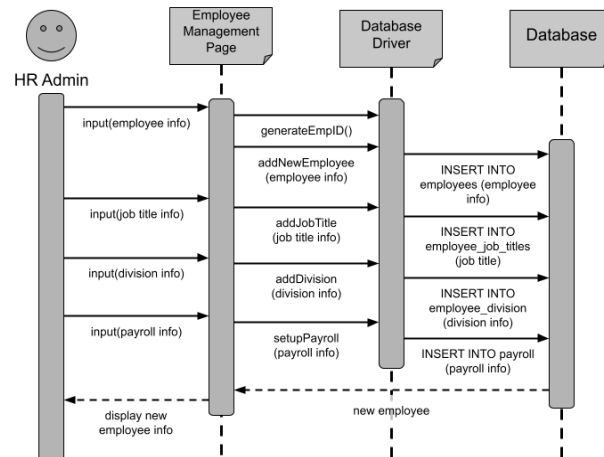
Employee Search Sequence Diagram



Update Salary by % Sequence Diagram



Add New Employee Sequence Diagram



## 7. APPENDIX

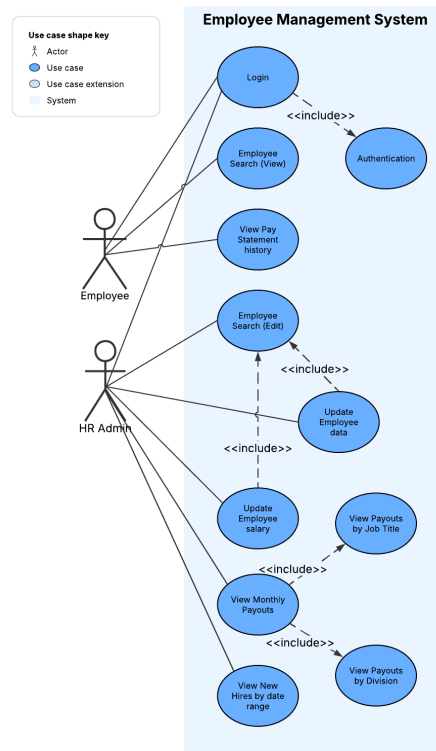
### 7.1. Definitions and Acronyms

EMS: Employee Management System

HR: Human Resources

JDBC: Java Database Connectivity

### 7.2. Additional UML diagrams



### 7.3. Reflection

This project gave us valuable hands-on experience building a complete software system using Java and MySQL. It helped strengthen our understanding of key concepts like database design, object-oriented programming, and overall system structure. Beyond the technical side, it also showed how important communication and teamwork are when working on a group project. This experience was useful in preparing us for real-world software development and gave a better idea of what to expect in a professional environment.